



The Inner Workings of *StarCraft II*: Using Statistics to Sharpen the Competitive Edge

Bayard Roberts
Mentor: Prof. Choong-Soo Lee
With assistance from Prof. Ivan Ramler
Department of Math, CS, and Statistics
St. Lawrence University

Introduction

- StarCraft II* is a 1v1, competitive, online real-time-strategy game.



- Players choose one of three races and fight each other for victory.
- Zerg is a race of creatures that can rapidly mutate into beasts with wings and claws.
- Terran is a race of humans that create armored suits, tanks, and war machines.
- Protoss is a race of advanced aliens that make use of futuristic technologies and psionic powers.
- In 2014, over \$35M was awarded in prize money for *StarCraft II* e-sport tournaments.



Methods & Materials

- Matches played as part of the *StarCraft II* World Championship Series (WCS) are recorded and made available for download.

StarCraft II Replays
• ~3500 Matches

SC2Reader
• Python library

Data extraction
program written
in Python

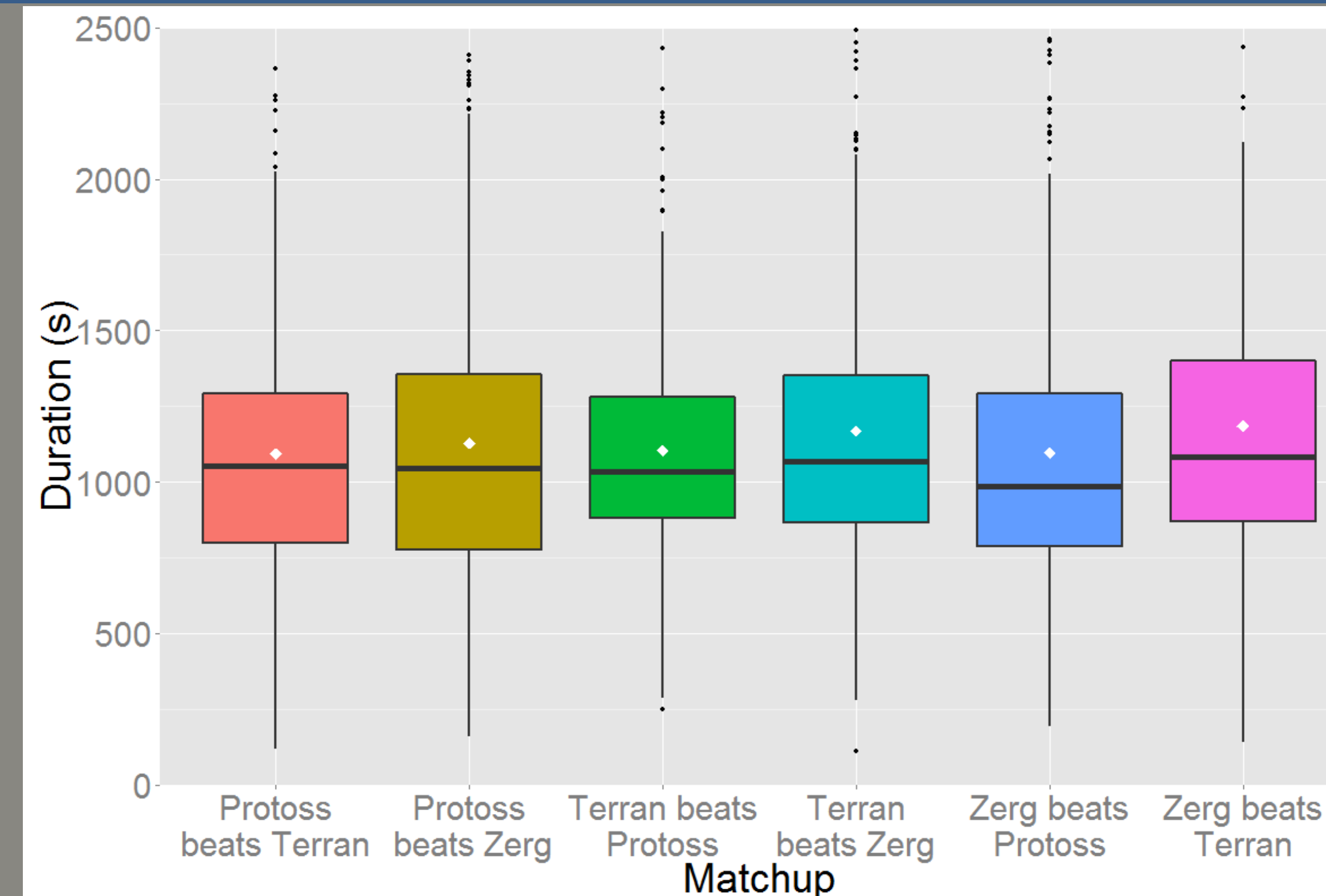
CSV file

R Analysis

Results

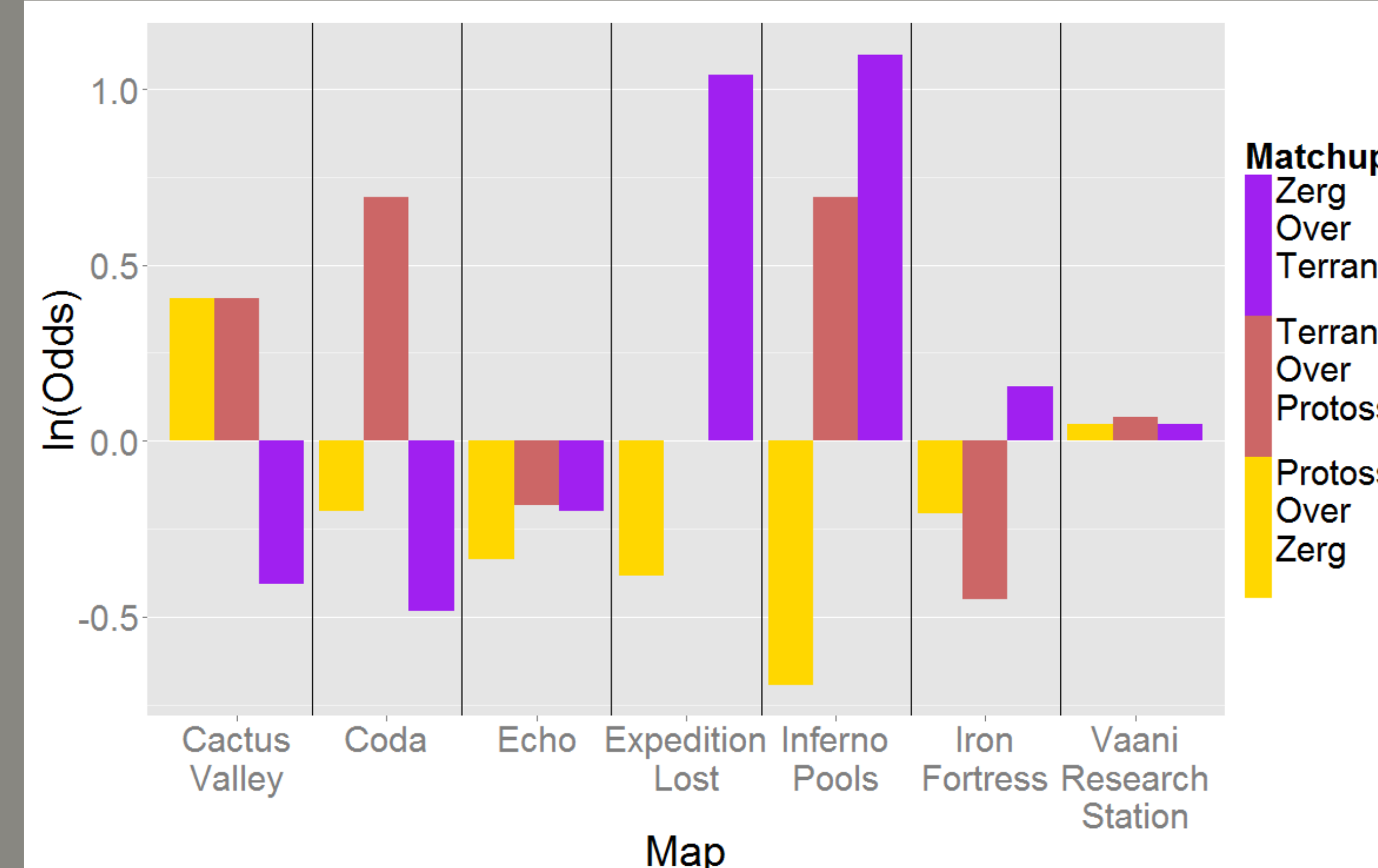
Game Duration

- Does game duration play a role in giving a certain race an advantage?
- Certain race matchups might do better in short games or longer games.
 - Zerg is especially known for being able to create a starting army faster than the other races.
- By looking at average game duration of victory matches for each race matchup, we can determine if there is evidence that game length helps or hinders certain races. The graph reveals that duration does not seem to give or take an advantage to any race in any matchup. Indeed, a significance test reveals that none of the mean durations are significantly different.



Map

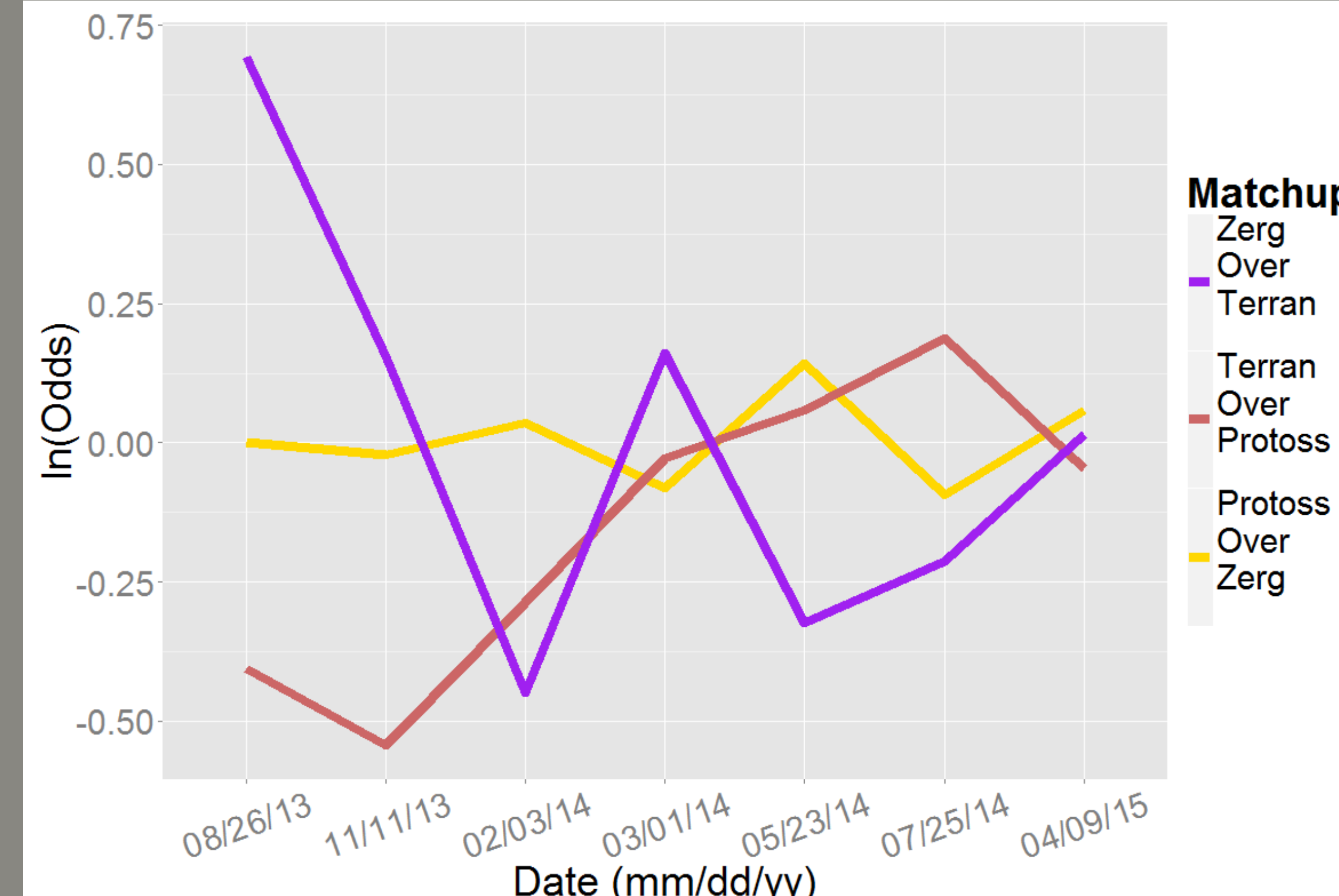
- Does the game map give an advantage toward any race matchup?
- Certain race matchups might do better in maps that are large or small, or in maps that have sections that only allow air units.
 - Zerg has the fastest standard land unit in the game. Large maps, like *Inferno Pools*, might favor the fast Zerg.
- By looking at the seven maps played in season 2 of WCS 2015, we can see if there is evidence that map choice can influence certain matchup victories. The graph first reveals that some maps do indeed favor or disfavor certain races and matchups. On the map *Expedition Lost*, Zerg beat Terran more than Terran beat Zerg, significant at a 5% level. This suggests that the Zerg Terran matchup on *Expedition Lost* favors Zerg. Other significant differences exist as well.



Bars above 0 favor the top race in the legend, while bars below 0 favor the bottom race. For example, when a yellow bar is negative, the Zerg tend to beat the Protoss on that map.

Game Patches

- Starcraft II* is changed every couple months in updates that are called game patches. Some patches change the balance of the game by changing the usefulness of individual units.
- Documentation of the changes to the game in each patch are available on the internet. By looking at the rise and fall of each matchup, we can see how effective specific changes were.
- The graph reveals that the game patches have, in general, caused each matchup to tend closer to the 0.0 mark, implying that matches do not favor a particular race. Early on in *Starcraft II*'s life, the Zerg-Terran and Terran-Protoss matchups might have been unfair, but the game has been changed to create more balance.



Further Results

- Analysis of individual variables has revealed that *StarCraft II* is a well balanced game. Most variables do little to predict outcome of games.
 - Game Duration
 - Maximum actions per minute, a measure of how fast a player can input commands
 - Average unspent resources. Resources that are left unspent do nothing to help victory. Minimizing them is considered a good thing, but it turned out to be insignificant.
- Race matchup without considering other variables does show some significant results, but any advantage seems dependent on what version of *StarCraft II* is being played.

For the Future

- I will be analyzing more ideas with Prof. Ivan Rambler.
 - Number of bases created and destroyed.
 - The differences of certain actions in early, mid, and late game.
 - Regression analysis to see the relationship of multiple variables.
- Starcraft II* is expected to be played competitively for a long while. This research can be built upon and prove useful years into the future.

References

WCS Replay Packs
<http://wcs.battle.net/sc2/en/articles/sc2reader>
<https://github.com/GraylinKim/sc2reader>

Acknowledgements

Funding for this project was provided by the Lorna A. Ness Fellows Fund, Daniel F. '65, and the Ann H. Sullivan Endowment for Student/Faculty Research.